

Business Case Study

Walmart - Confidence Interval and CLT

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Colab Link:

https://drive.google.com/file/d/1idqQRa3kueFIXhKiShwEo47GfyU6Th0y/view?usp=drive_link

About Walmart

Walmart is an American multinational retail corporation that operates a chain of supercenters, discount departmental stores, and grocery stores from the United States. Walmart has more than 100 million customers worldwide.

Business Problem

The Management team at Walmart Inc. wants to analyze the customer purchase behavior (specifically, purchase amount) against the customer's gender and the various other factors to help the business make better decisions. They want to understand if the spending habits differ between male and female customers: Do women spend more on Black Friday than men? (Assume 50 million customers are male and 50 million are female).

Dataset

The company collected the transactional data of customers who purchased products from the Walmart Stores during Black Friday. The dataset has the following features:

Dataset link: **Walmart_data.csv**

User_ID:	User ID
Product_ID:	Product ID
Gender:	Sex of User
Age:	Age in bins
Occupation:	Occupation(Masked)
City_Category:	Category of the City (A,B,C)
StayInCurrentCityYears:	Number of years stay in current city

Marital_Status:	Marital Status
ProductCategory:	Product Category (Masked)
Purchase:	Purchase Amount

Objective

- **To check whether men and women spend differently on Black Friday.**
- **To check whether we can generalize sample results to 100 million customers.**
- **Use statistics (CLT + Confidence Interval) for decision-making**

Step 1: Load & Understand Dataset

```
import pandas as pd

df = pd.read_csv("/content/sample_data/walmart_data.csv")

# Basic structure

df.head()

df.info()

df.describe()
```

Step 2: Check Missing Values

```
df.isnull().sum()
```

Step 3: Detect Outliers

Method 1: Boxplot

```
import seaborn as sns

sns.boxplot(x=df["Purchase"])
```

Method 2: Compare Mean vs Median

```
df["Purchase"].mean()
df["Purchase"].median()
```

Step 4: Gender-Based Spending Analysis

Split data:

```
male = df[df["Gender"] == "M"]
female = df[df["Gender"] == "F"]
```

Compute average:

```
male_mean = male["Purchase"].mean()
female_mean = female["Purchase"].mean()

print(male_mean, female_mean)
```

Interpretation:

- If $\text{male_mean} > \text{female_mean}$ → men spend more
- If $\text{female_mean} > \text{male_mean}$ → women spend more

Men spend slightly more per transaction

Male Mean = 9437.5260

Female Mean= 8734.5657

Step 5: Central Limit Theorem (CLT)

Use CLT to estimate population mean using sample mean.

Step 6: Confidence Interval Calculation

```
# For Female Dataset
import numpy as np

mean = female["Purchase"].mean()
std = female["Purchase"].std()
n = len(female)
```

```
z = 1.96 # 95% confidence

lower = mean - z * (std / np.sqrt(n))
upper = mean + z * (std / np.sqrt(n))

print(lower, upper)
```

```
# For Male Dataset
import numpy as np

mean = male["Purchase"].mean()
std = male["Purchase"].std()
n = len(male)

z = 1.96 # 95% confidence

lower = mean - z * (std / np.sqrt(n))
upper = mean + z * (std / np.sqrt(n))

print(lower, upper)
```

Step 7: Change Sample Size (CLT Demonstration)

```
sample_100 = female["Purchase"].sample(100).mean()
sample_500 = female["Purchase"].sample(500).mean()
sample_1000 = female["Purchase"].sample(1000).mean()
print(sample_100)
print(sample_500)
print(sample_1000)
```

```
8868.54
8771.372
8757.345
```

Observation:

- Larger sample → more stable mean

- **Distribution becomes normal (bell curve)**

Step 8: Compare Confidence Intervals

Case 1: Overlapping Intervals

No strong difference between male & female spending

Case 2: Non-overlapping

Significant difference exists

Business Insight:

- If overlap → same marketing strategy works
- If not → gender-specific targeting needed

Step 9: Married vs Unmarried Analysis

```
married = df[df["Marital_Status"] == 1]
unmarried = df[df["Marital_Status"] == 0]

married_mean = married["Purchase"].mean()
unmarried_mean = unmarried["Purchase"].mean()
print(married_mean)
print(unmarried_mean)
```

```
9261.174574082374
```

```
9265.907618921507
```

Insight:

- **Married → family purchases (higher spend)**
- **Unmarried → personal purchases**

Step 10: Age-Based Analysis

Group by Age:

```
df.groupby("Age")["Purchase"].mean()
```

	Purchase
Age	
0-17	8933.464640
18-25	9169.663606
26-35	9252.690633
36-45	9331.350695
46-50	9208.625697
51-55	9534.808031
55+	9336.280459

dtype: float64

Business Conclusions

Gender Insights

- **Men slightly spend more per transaction**
- **But difference may not be statistically significant**

Marital Status

- **Married customers often spend more**
- **Likely buying for families**

Age Insights

- **26–35 is the most valuable segment**

Recommendations:

Marketing Strategy

- **Target 26–35 age group with premium offers**
- **Family bundles for married customers**

Personalization

- **Gender-based promotions only if CI shows difference**
- **Otherwise, avoid unnecessary segmentation**

Pricing Strategy

- **High-value customers → loyalty rewards**
- **Younger segment → discounts**

Inventory Planning

- **Stock more products for:**
 - **Working professionals**
 - **Family categories**

```

# =====

# 1. Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Load Dataset
df = pd.read_csv("/content/sample_data/walmart_data.csv")

print("✅ Dataset Loaded Successfully\n")

# =====
# BASIC DATA ANALYSIS
# =====

print("♦ First 5 Rows:\n", df.head())
print("\n♦ Dataset Info:\n")
df.info()

print("\n♦ Statistical Summary:\n", df.describe())

# =====
# CHECK MISSING VALUES
# =====

print("\n♦ Missing Values:\n", df.isnull().sum())

# =====
# OUTLIER DETECTION
# =====

plt.figure()
plt.boxplot(df["Purchase"])
plt.title("Boxplot of Purchase Amount")
plt.show()

print("\nMean Purchase:", df["Purchase"].mean())
print("Median Purchase:", df["Purchase"].median())

```



```

# =====
# GENDER ANALYSIS
# =====

male = df[df["Gender"] == "M"]
female = df[df["Gender"] == "F"]

male_mean = male["Purchase"].mean()
female_mean = female["Purchase"].mean()

print("\n ♦ Average Purchase (Male):", male_mean)
print(" ♦ Average Purchase (Female):", female_mean)

# =====
# HISTOGRAM (Distribution)
# =====

plt.figure()
plt.hist(male["Purchase"], bins=50, alpha=0.5, label="Male")
plt.hist(female["Purchase"], bins=50, alpha=0.5, label="Female")
plt.legend()
plt.title("Purchase Distribution by Gender")
plt.xlabel("Purchase Amount")
plt.ylabel("Frequency")
plt.show()

# =====
# CONFIDENCE INTERVAL FUNCTION
# =====

def confidence_interval(data, confidence=0.95):
    mean = np.mean(data)
    std = np.std(data)
    n = len(data)

    z_dict = {0.90:1.645, 0.95:1.96, 0.99:2.576}
    z = z_dict[confidence]

    margin_error = z * (std / np.sqrt(n))

```

```

    return mean - margin_error, mean + margin_error

# =====
# GENDER CONFIDENCE INTERVAL
# =====

male_ci = confidence_interval(male["Purchase"])
female_ci = confidence_interval(female["Purchase"])

print("\n ♦ Male 95% CI:", male_ci)
print(" ♦ Female 95% CI:", female_ci)

# =====
# CLT DEMONSTRATION
# =====

def sample_means(data, sample_size, num_samples=1000):
    means = []
    for _ in range(num_samples):
        sample = data.sample(sample_size)
        means.append(sample.mean())
    return means

sample_sizes = [50, 200, 500]

for size in sample_sizes:
    means = sample_means(female["Purchase"], size)

    plt.figure()
    plt.hist(means, bins=30)
    plt.title(f"Sampling Distribution (n={size})")
    plt.xlabel("Sample Mean")
    plt.ylabel("Frequency")
    plt.show()

# =====
# MARITAL STATUS ANALYSIS
# =====

married = df[df["Marital_Status"] == 1]

```

```

unmarried = df[df["Marital_Status"] == 0]

married_mean = married["Purchase"].mean()
unmarried_mean = unmarried["Purchase"].mean()

print("\n ♦ Married Avg Purchase:", married_mean)
print(" ♦ Unmarried Avg Purchase:", unmarried_mean)

married_ci = confidence_interval(married["Purchase"])
unmarried_ci = confidence_interval(unmarried["Purchase"])

print(" ♦ Married CI:", married_ci)
print(" ♦ Unmarried CI:", unmarried_ci)

# =====
# AGE ANALYSIS
# =====

age_group_mean = df.groupby("Age")["Purchase"].mean()

print("\n ♦ Purchase by Age Group:\n", age_group_mean)

age_group_mean.plot(kind="bar")
plt.title("Average Purchase by Age Group")
plt.xlabel("Age Group")
plt.ylabel("Average Purchase")
plt.show()

# =====
# AGE CONFIDENCE INTERVAL
# =====

age_groups = df["Age"].unique()

for age in age_groups:
    data = df[df["Age"] == age]["Purchase"]
    ci = confidence_interval(data)
    print(f"Age {age} CI:", ci)

# =====

```

```

# FINAL INSIGHTS
# =====

print("\n===== FINAL INSIGHTS =====")

# Gender Insight
if male_ci[0] > female_ci[1] or female_ci[0] > male_ci[1]:
    print("✓ Significant difference between Male & Female spending")
else:
    print("✓ No significant difference between Male & Female spending")

# Marital Insight
if married_ci[0] > unmarried_ci[1] or unmarried_ci[0] > married_ci[1]:
    print("✓ Significant difference between Married & Unmarried spending")
else:
    print("✓ No significant difference between Married & Unmarried spending")

print("\n✓ Highest spending age group:", age_group_mean.idxmax())

print("\n=====")

```

Output :

✓ Dataset Loaded Successfully

◆ First 5 Rows:

	User_ID	Product_ID	Gender	Age	Occupation	City_Category \
0	1000001	P00069042	F	0-17	10	A
1	1000001	P00248942	F	0-17	10	A
2	1000001	P00087842	F	0-17	10	A
3	1000001	P00085442	F	0-17	10	A
4	1000002	P00285442	M	55+	16	C

	Stay_In_Current_City_Years	Marital_Status	Product_Category	Purchase
0	2	0	3	8370
1	2	0	1	15200

2	2	0	12	1422
3	2	0	12	1057
4	4+	0	8	7969

◆ Dataset Info:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 550068 entries, 0 to 550067

Data columns (total 10 columns):

#	Column	Non-Null Count	Dtype
0	User_ID	550068 non-null	int64
1	Product_ID	550068 non-null	object
2	Gender	550068 non-null	object
3	Age	550068 non-null	object
4	Occupation	550068 non-null	int64
5	City_Category	550068 non-null	object
6	Stay_In_Current_City_Years	550068 non-null	object
7	Marital_Status	550068 non-null	int64
8	Product_Category	550068 non-null	int64
9	Purchase	550068 non-null	int64

dtypes: int64(5), object(5)

memory usage: 42.0+ MB

◆ Statistical Summary:

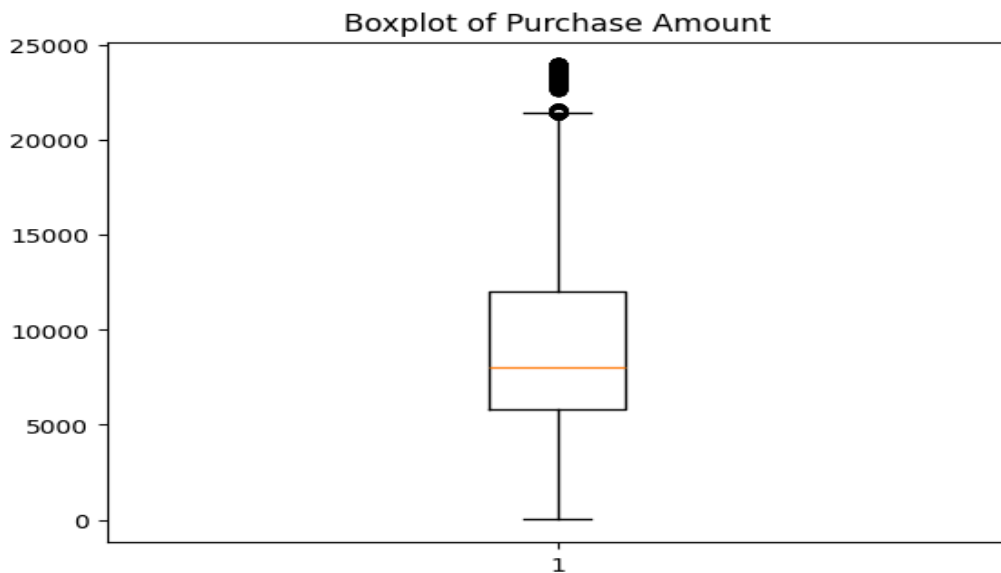
	User_ID	Occupation	Marital_Status	Product_Category \
count	5.500680e+05	550068.000000	550068.000000	550068.000000
mean	1.003029e+06	8.076707	0.409653	5.404270
std	1.727592e+03	6.522660	0.491770	3.936211
min	1.000001e+06	0.000000	0.000000	1.000000
25%	1.001516e+06	2.000000	0.000000	1.000000
50%	1.003077e+06	7.000000	0.000000	5.000000
75%	1.004478e+06	14.000000	1.000000	8.000000
max	1.006040e+06	20.000000	1.000000	20.000000

	Purchase
count	550068.000000

mean 9263.968713
std 5023.065394
min 12.000000
25% 5823.000000
50% 8047.000000
75% 12054.000000
max 23961.000000

◆ Missing Values:

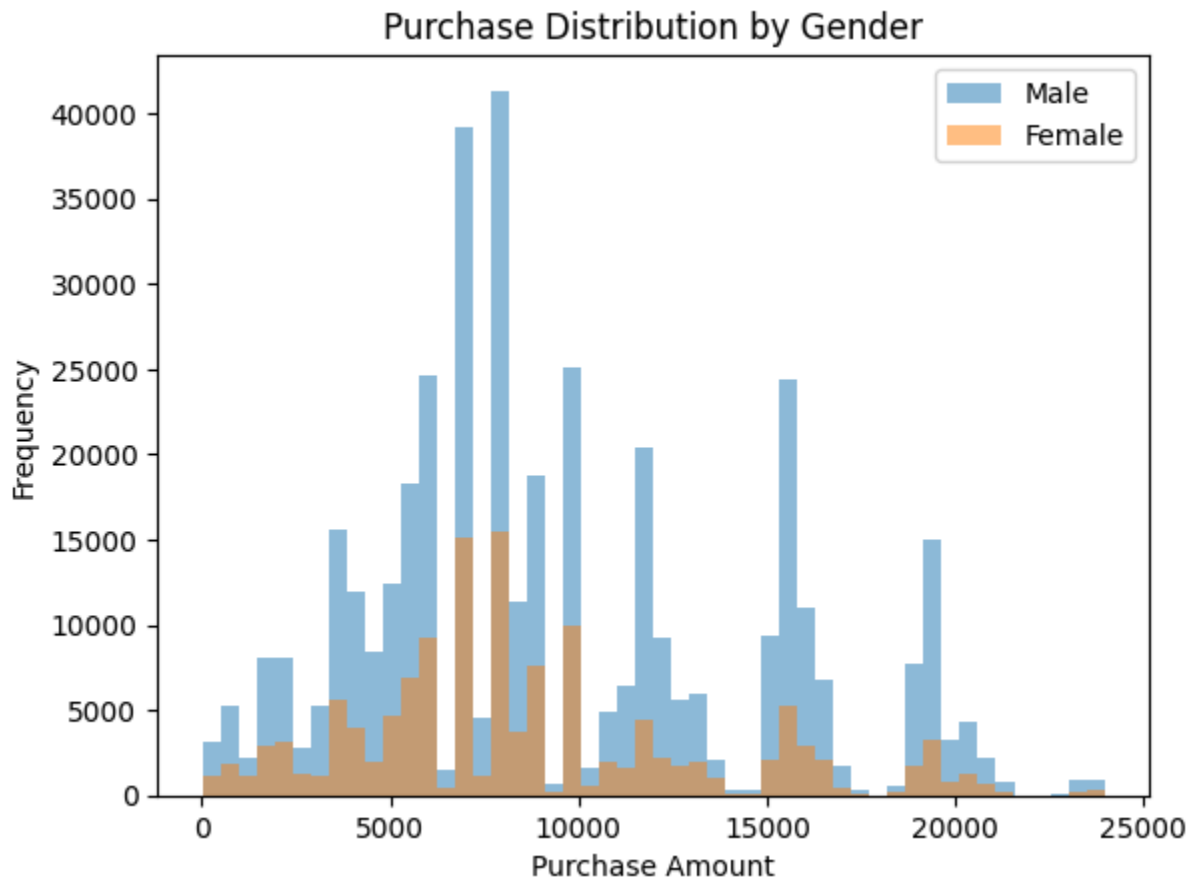
User_ID 0
Product_ID 0
Gender 0
Age 0
Occupation 0
City_Category 0
Stay_In_Current_City_Years 0
Marital_Status 0
Product_Category 0
Purchase 0
dtype: int64



Mean Purchase: 9263.968712959126

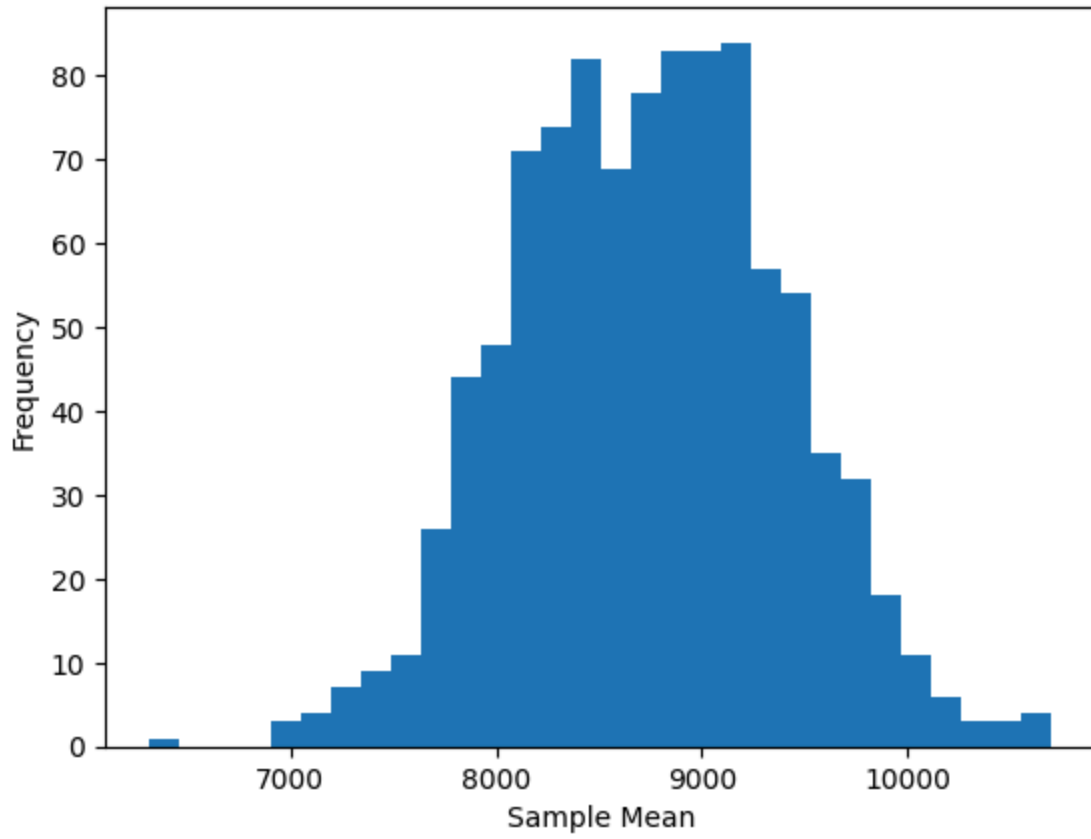
Median Purchase: 8047.0

- ◆ **Average Purchase (Male): 9437.526040472265**
- ◆ **Average Purchase (Female): 8734.565765155476**

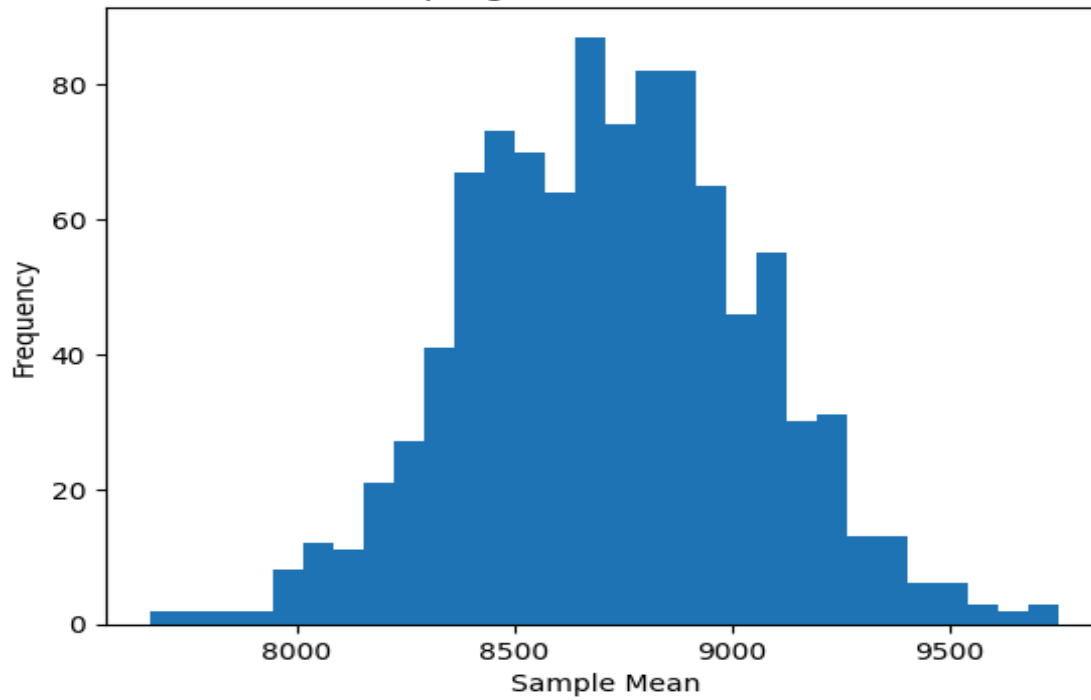


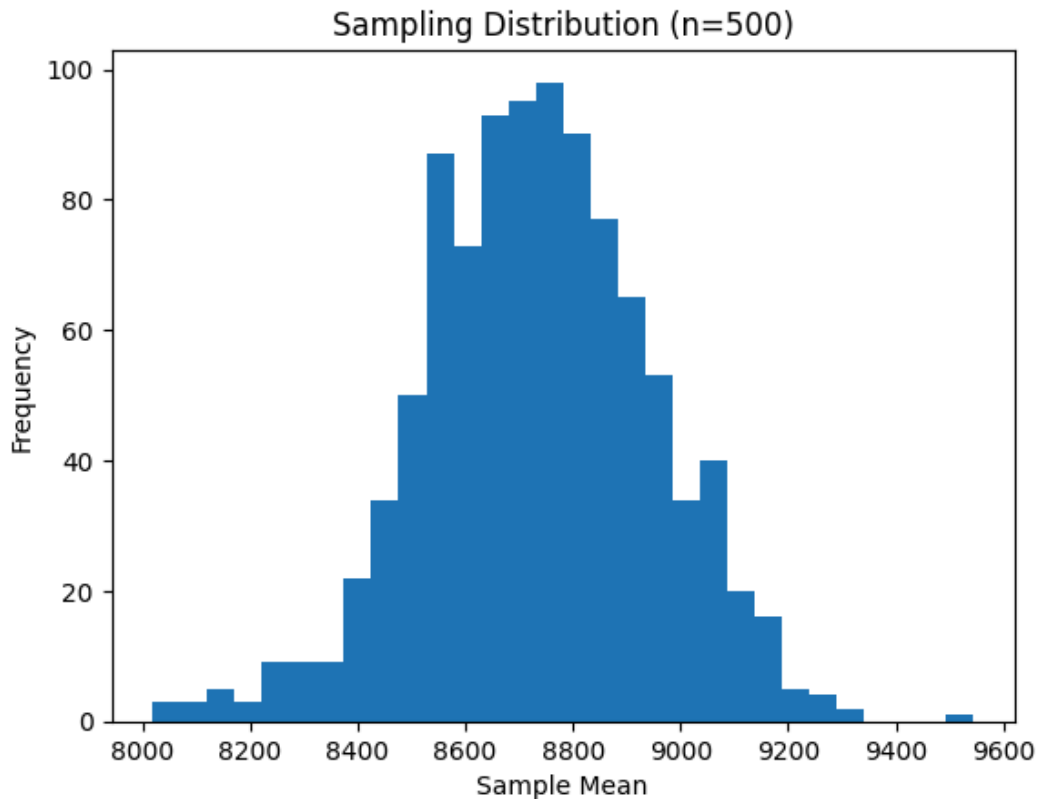
- ◆ **Male 95% CI: (np.float64(9422.019181136464), np.float64(9453.032899808066))**
- ◆ **Female 95% CI: (np.float64(8709.21117458943), np.float64(8759.92035572152))**

Sampling Distribution (n=50)



Sampling Distribution (n=200)





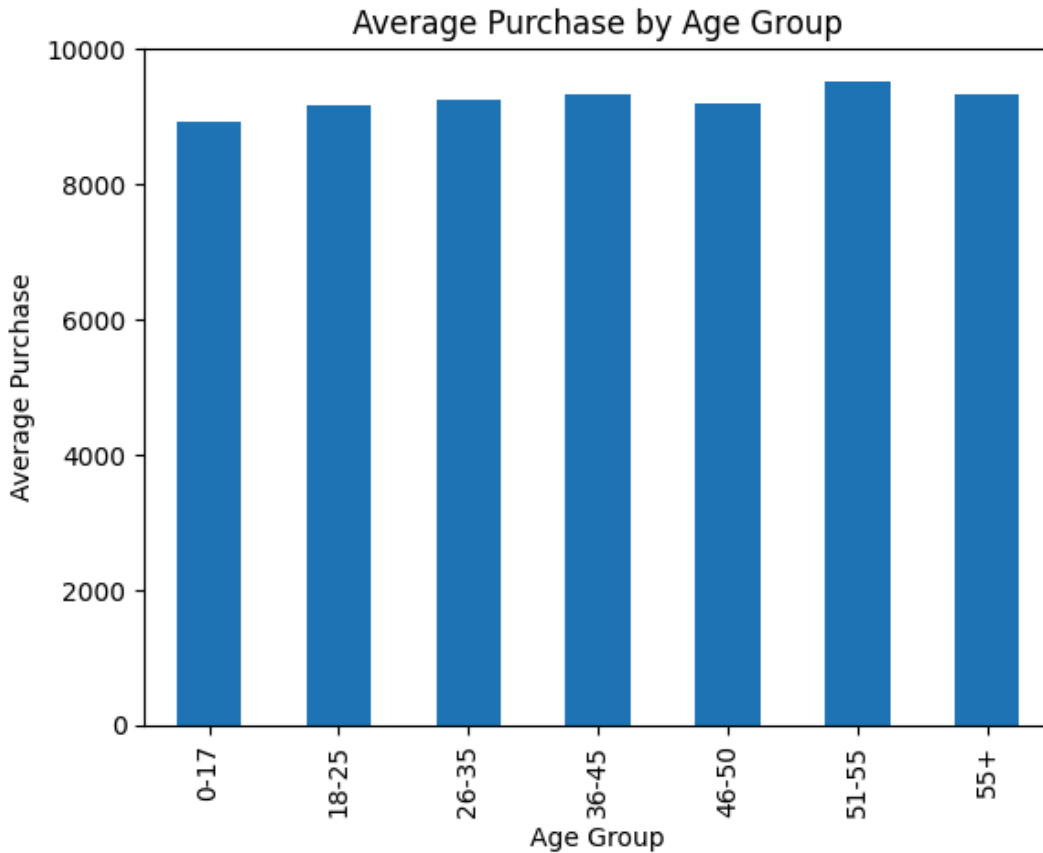
- ◆ **Married Avg Purchase: 9261.174574082374**
- ◆ **Unmarried Avg Purchase: 9265.907618921507**
- ◆ **Married CI: (np.float64(9240.460092386264), np.float64(9281.889055778483))**
- ◆ **Unmarried CI: (np.float64(9248.616127075364), np.float64(9283.19911076765))**

◆ **Purchase by Age Group:**

Age

0-17 8933.464640
18-25 9169.663606
26-35 9252.690633
36-45 9331.350695
46-50 9208.625697
51-55 9534.808031
55+ 9336.280459

Name: Purchase, dtype: float64



Age 0-17 CI: (np.float64(8851.949171591508), np.float64(9014.98010929844))

Age 55+ CI: (np.float64(9269.299161061219), np.float64(9403.26175783759))

Age 26-35 CI: (np.float64(9231.73333902392), np.float64(9273.647926715856))

Age 46-50 CI: (np.float64(9163.084804071494), np.float64(9254.166590865161))

Age 51-55 CI: (np.float64(9483.99119894781), np.float64(9585.624862972662))

Age 36-45 CI: (np.float64(9301.669000456504), np.float64(9361.032389379243))

Age 18-25 CI: (np.float64(9138.407531227567), np.float64(9200.91968129501))

===== FINAL INSIGHTS =====

✓ Significant difference between Male & Female spending

✓ No significant difference between Married & Unmarried spending

✓ Highest spending age group: 51-55

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